



IILM UNIVERSITY

Greater Noida, Uttar Pradesh | School of Computer Science & Engineering

Automation Testing of SauceDemo Website Using Selenium with Python

STUDENT NAME

Ankit Kumar Dubey

ENROLLMENT / ROLL NO.

2410031285

PROGRAM & SEMESTER

B.Tech CSE — 5th Semester (Batch 2024–28)

ORGANIZATION

Sopra Steria Group

Introduction & Internship Objective

Software testing is a critical phase of the Software Development Life Cycle (SDLC), verifying that an application behaves as expected. As applications grow more complex and release cycles shorten, automation testing has become essential — improving speed, accuracy and consistency over manual testing.

This internship focused on automating the testing of the SauceDemo e-commerce website using Selenium WebDriver with Python, built around the Page Object Model (POM) to improve readability, maintainability and future scalability.

AT A GLANCE

- **Application Under Test**

SauceDemo e-commerce website

- **Automation Tool**

Selenium WebDriver

- **Language**

Python

- **Design Pattern**

Page Object Model (POM)

- **Host Organization**

Sopra Steria Group

Organization Profile — Sopra Steria

Sopra Steria is one of Europe's leading technology consulting and digital transformation organizations, delivering end-to-end business solutions across consulting, software development, systems integration, cloud, cybersecurity, data analytics, AI and quality engineering.

It operates across multiple countries, serving clients in banking, insurance, aerospace, defence, government, transportation, retail, manufacturing and telecommunications — helping businesses accelerate digital transformation.

Hands-on exposure to real-world QA processes and POM-based framework design during the internship significantly strengthened technical, analytical and debugging skills.

Global Reach

End-to-end technology consulting across multiple countries

Multi-Industry

Banking, insurance, aerospace, defence, retail & more

Innovation Culture

Continuous learning, teamwork and technical excellence

Problem Statement & Objectives

PROBLEM STATEMENT

Manual testing of a web application such as SauceDemo is repetitive, time-consuming and prone to human error when core flows — login, product verification, sorting, cart, checkout and logout — must be repeated across every release. A reliable, repeatable automation mechanism was needed to validate these flows without manual intervention.

PROJECT OBJECTIVES

Automate functional testing of the SauceDemo application using Selenium WebDriver with Python, eliminating repetitive manual effort and validating these core functionalities:

- User authentication — valid and invalid login scenarios

- Product sorting / filtering options

- Checkout process and order completion

- Product verification and inventory inspection

- Shopping cart operations

- Logout functionality

Scope of the Project & Technologies Used

SCOPE — CORE FEATURES COVERED

- Login validation (successful & invalid)
- Product information & inventory verification
- Sorting: A–Z, Z–A, Price low–high, high–low
- Shopping cart operations & content checks
- Checkout: customer info & order overview
- Order completion & confirmation validation
- Application state reset
- Logout functionality & verification

Future extension: cross-browser automation, data-driven testing (Excel/CSV), parallel execution, reporting tools and CI/CD integration.

TECHNOLOGIES & TOOLS

Software	Purpose
Python 3.x	Programming language
Selenium WebDriver	Browser automation tool
Visual Studio Code	Development environment (IDE)
Google Chrome	Target web browser
ChromeDriver	Browser driver bridge

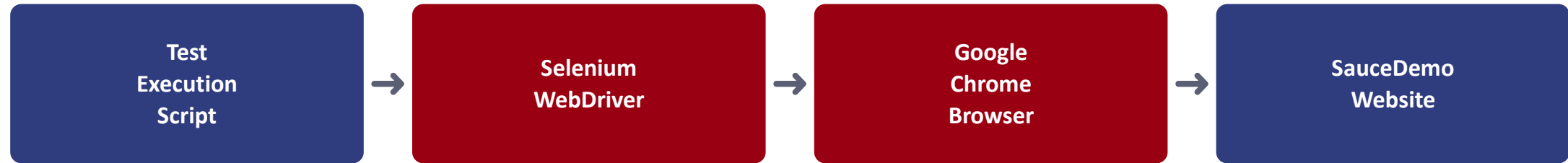
MINIMUM HARDWARE

Processor	Intel Core i5 or higher
RAM	8 GB minimum (16 GB recommended)
OS	Windows 11

PROJECT DESCRIPTION

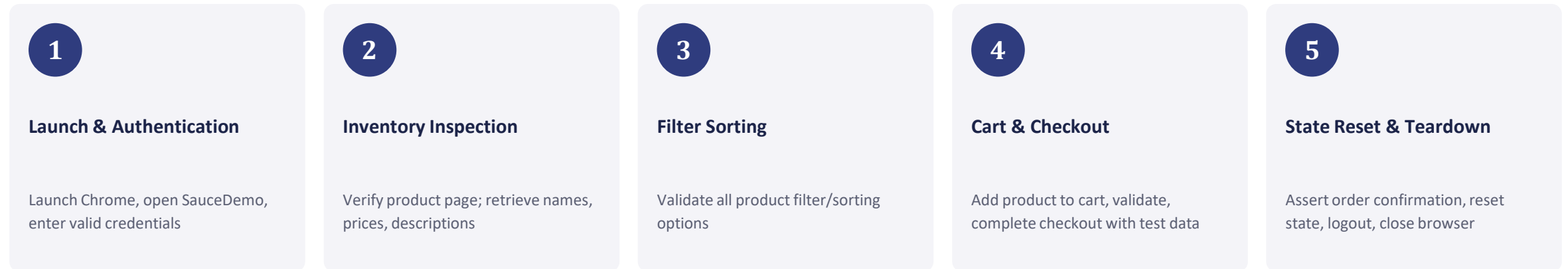
System Architecture & Methodology

LAYERED ARCHITECTURE — PAGE OBJECT MODEL (POM)



POM Layer sits between tests and the browser: Login • Inventory • Cart • Checkout • Menu page classes

METHODOLOGY — EXECUTION FLOW



Module Implementation Walkthrough



Login Module

Valid credential authentication into Inventory page



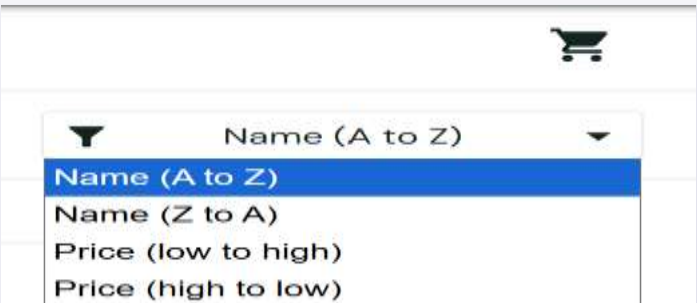
Invalid Login Module

Error message shown; unauthorized access prevented



Inventory Module

Product names, prices & descriptions extracted



Product Filter Module

Name/Price sorting validated (A–Z, Z–A, Low–High, High–Low)



Cart Module

Cart contents matched against selected product data



Checkout & Order Completion

Customer info entered; order confirmation asserted

Assertions Used & Expected Outcomes

KEY VALIDATION CHECKS

Login Validation

URL/title contains “inventory.html” after valid login

Cart Validation

Cart badge count equals number of chosen products

Cart Verification

Name, price, description & quantity match exactly

Checkout Validation

Flow advances into the Checkout Overview page

Order Completion

“THANK YOU FOR YOUR ORDER” header is displayed

Logout Validation

User is returned to the root login screen

EXPECTED OUTCOMES

The automation framework successfully executed the full end-to-end workflow without manual intervention — authenticating users, catching invalid logins, validating filters, completing purchases, confirming order completion, and logging out cleanly.

Every assertion in the test scripts passed successfully — demonstrating that automation cuts execution time to a fraction of manual overhead.

- Regression Testing
- Smoke Testing
- CI/CD Integration
- Efficiency Gains

Challenges Faced & Future Enhancements

CHALLENGES FACED

Browser Driver Configuration

Resolved by matching ChromeDriver version exactly to installed Chrome

Element Identification

Resolved using fixed IDs, structural XPaths & robust CSS selectors

Synchronization Issues

Resolved with precise Selenium dynamic wait strategies

Project Organization

Resolved by strictly following the Page Object Model (POM)

FUTURE ENHANCEMENTS

- Advanced reporting: PyTest HTML, Allure or Extent Reports with error screenshots
- Data-driven testing via external Excel/CSV file integration
- Multi-browser parallel execution (Chrome, Firefox, Safari, Edge) via Selenium Grid
- Full CI/CD pipeline orchestration using GitHub Actions or Jenkins

CONCLUSION

A Complete, Reliable Automation Framework

The automation testing of the SauceDemo website using Selenium WebDriver with Python was successfully completed during the internship at Sopra Steria Group. The framework demonstrated how automated scripts reduce repetitive overhead while improving test accuracy across multiple operational workflows. Following the modular Page Object Model (POM) pattern keeps the codebase readable and ready for future updates — and the real-world quality assurance skills gained provide a strong foundation for professional growth.

100%

Assertions passed across all test scripts

8

Core functional areas automated end-to-end

POM

Framework design pattern for scalability



THANK YOU

Ankit Kumar Dubey | 2410031285 | B.Tech CSE, 5th Semester | IILM University